

HORMONE PRODUCERS

HORMONES CARRY THE CHEMICAL DATA THAT CONTROL THE RATE AT WHICH GLANDS AND OTHER ORGANS WORK. HORMONE-PRODUCING CELLS ARE FOUND ALL AROUND THE BODY, MANY IN GLANDS THAT HAVE SPECIALIZED FUNCTIONS.

MASTER GLAND: THE PITUITARY

The pituitary, or hypophysis, is the most influential gland in the endocrine system. It comprises two distinct glands in one. The front (anterior) lobe, also known as the adenohypophysis, forms the larger part. Behind is the posterior lobe, or neurohypophysis. The anterior pituitary manufactures eight major hormones on

site and releases them into the bloodstream. The posterior pituitary receives its two main hormones from the hypothalamus, which lies above it; there, they are made by neurosecretory cells. Other neurosecretory cells make regulatory hormones, which travel via capillaries to the anterior lobe and control the release of hormones there.

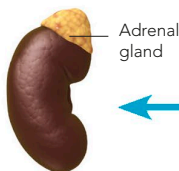
SKIN

The action of melanocyte-stimulating hormone (MSH) – produced in a thin layer between the two pituitary lobes – causes cells called melanocytes in skin tissue to produce more melanin pigment, making the skin darken.



ADRENAL GLAND

Adrenocorticotrophic hormone (ACTH) triggers the adrenals to produce steroid hormones that control stress response and the body's use of fats, carbohydrates, proteins, and minerals.



THYROID

Thyrotropin-releasing hormone from the hypothalamus controls the release of thyroid-stimulating hormone (TSH). This encourages the thyroid to become more active and affects metabolism.



BONE AND GENERAL GROWTH

Growth hormone (GH) acts on the whole body to promote protein manufacture, bone growth, and building of new tissues throughout life, but is especially important for development in children.

